

Real-Time 3-D Object Detection for Autonomous Driving Using Sparse Convolutional Neural Networks

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DOI: [10.1109/TITS.2025.3141872](https://doi.org/10.1109/TITS.2025.3141872)

Abstract

Accurate and low-latency 3-D object detection from LiDAR point clouds is a prerequisite for safe autonomous vehicle (AV) operation. Current voxel-based methods achieve high mean average precision (mAP) on benchmark datasets but suffer from excessive GPU memory usage that prevents real-time inference on embedded automotive-grade hardware. We present SparseDet3D, a sparse convolutional architecture that exploits the inherent sparsity of outdoor LiDAR scans to reduce memory consumption by 67% and inference latency by 41% relative to VoxelNet, while maintaining 78.4 mAP on the Waymo Open Dataset 3-D detection benchmark. Our approach introduces a dynamic voxel grouping strategy that adapts receptive field size to object density, improving recall for pedestrians and cyclists by 12.3 percentage points. We validate SparseDet3D on an NVIDIA Orin SoC running at 30 Hz without quantisation, meeting the latency budget required for ISO 26262 ASIL-D safety classification. A production deployment study across 2.1 million kilometres of road testing in mixed urban and highway scenarios confirms a false-positive rate below 1.2×10^{-4} detections per kilometre, a critical metric for regulatory approval.

Keywords: LiDAR point cloud; 3-D object detection; sparse convolution; autonomous driving; real-time inference; ASIL

1. Introduction

The commercialisation of SAE Level 4 autonomous vehicles requires perception systems that reliably detect and classify surrounding objects—vehicles, pedestrians, cyclists, and infrastructure elements—at highway speeds with end-to-end latencies below 33 ms. LiDAR sensors provide millimetre-precise depth measurements and are robust to changes in illumination, making them the primary input modality for perception in production AV stacks deployed by Waymo, Cruise, and Baidu Apollo. However, processing the 100,000–200,000 points generated per LiDAR sweep in real time on constrained embedded processors remains an open engineering problem.

Voxel-based methods such as VoxelNet and PointPillars partition the 3-D space into a regular grid and apply 3-D or pseudo-2-D convolutions to extract spatial features. Their accuracy scales with voxel resolution, but at resolutions sufficient for pedestrian detection the memory footprint of the 3-D feature maps exceeds 16 GB, precluding deployment on automotive-grade GPUs with 8–12 GB VRAM. Existing sparsification techniques reduce computation but introduce irregular memory access patterns that degrade throughput on production inference engines.

In this paper we make the following contributions: (1) SparseDet3D, a fully sparse convolutional backbone that eliminates dense feature map materialisation; (2) a dynamic voxel grouping algorithm that increases effective resolution in dense regions without proportional memory cost; (3) a

comprehensive ablation study on Waymo Open Dataset and nuScenes; and (4) end-to-end profiling on NVIDIA Orin at ISO 26262 ASIL-D latency constraints.

2. Methods

SparseDet3D processes raw LiDAR point clouds through four stages: dynamic voxelisation, sparse 3-D backbone, multi-scale feature pyramid, and anchor-free detection head. During dynamic voxelisation each point is assigned to the finest non-empty voxel at 5 cm resolution, and voxels containing more than $k = 12$ points are recursively split at 2.5 cm, creating an adaptive octree-style representation. The sparse backbone implements six SparseCNA blocks, each consisting of a $3 \times 3 \times 3$ sparse convolution, batch normalisation, and ReLU activation. Submanifold sparse convolutions are used in the first three blocks to preserve the active site mask, while stride-2 generalised sparse convolutions progressively downsample the feature volume. Skip connections aggregate multi-scale representations in a feature pyramid network (FPN) without converting to dense tensors.

The detection head follows a centre-point paradigm. For each object class $c \in \{\text{Car, Pedestrian, Cyclist, Motorcycle, Truck}\}$, a separate Gaussian heatmap head predicts the probability of an object centre at each pillar location. Box regression branches output $(\Delta x, \Delta y, \Delta z, l, w, h, \theta)$ in the sensor frame. Non-maximum suppression (NMS) is replaced by a faster peak-extraction operation on the GPU without CPU round-trips, reducing post-processing latency from 4.2 ms to 0.7 ms.

3. Results

Table 1 compares SparseDet3D with state-of-the-art methods on the Waymo Open Dataset Level-2 3-D detection task. SparseDet3D achieves 78.4 LEVEL_2 mAP averaged across all classes, compared with 77.1 for CenterPoint-Pillar and 76.8 for VoxelNeXt. GPU memory at inference time is 3.8 GB, 67% lower than the next best method. End-to-end latency on NVIDIA Orin at FP16 precision is 22.4 ms per frame, comfortably within the 33 ms budget.

Pedestrian recall at IoU 0.5 improves from 65.2% (VoxelNeXt) to 77.5% (SparseDet3D), driven by the dynamic voxel splitting that increases effective resolution around small objects. In the production road-testing programme, SparseDet3D operated continuously across 2.1 million km in 14 cities, recording a false-positive rate of 1.18×10^{-6} per km, below the 2.0×10^{-6} threshold required for ASIL-D certification under the UL 4600 standard.

4. Discussion and Commercial Implications

The 41% reduction in inference latency and 67% reduction in memory footprint enable SparseDet3D to run on the NVIDIA Orin at 30 Hz without quantisation-induced accuracy loss, removing a key engineering barrier to robotaxi deployment at scale. Tier-1 automotive suppliers have evaluated the architecture for integration into next-generation perception ECUs targeting model-year 2027 vehicles. Waymo has initiated an internal technology licensing discussion covering the dynamic voxel grouping patent (US provisional application 63/841,224). The projected unit cost reduction from eliminating the discrete LiDAR co-processor is USD 380 per vehicle, relevant at the expected robotaxi production volume of 100,000 units/year.

5. Conclusion

SparseDet3D demonstrates that sparse 3-D convolution architectures can achieve both state-of-the-art accuracy and production-grade efficiency for autonomous vehicle perception. Future work will explore sensor fusion with camera streams and radar to maintain performance under adverse weather conditions, a prerequisite for nationwide AV deployment beyond geofenced operational design domains.

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